

전자소자개론(공)(반) (SM303-01)

학기	2026년 1학기
담당교수	이병문
이수구분	심화필수
학점	3.0
강의시간	화10:30-12:00(E7 - 224), 목10:30-12:00(E7 - 224)
강의실	E3 312 (교수실)
소속	반도체공학과

교과개요

For the last several decades, microelectronics technology has shown tremendous innovation and improvements. This important technological innovation came along with the improvement of the development of electronic devices. It enabled state-of-the-art mobile computing devices, allowing analog/digital logic circuit implementation and signal processing, optoelectronic devices and sensors, and photovoltaic cells. In this course, first, we will briefly review the fundamental physics governing the semiconductor-based electronic device operations and fabrications: the structure of solids, the behavior of electrons in periodic lattice structures, semiconductor materials, etc. After that, we will study the structure, operation principle, and electrical characteristics as circuit elements of representative electronic devices such as diodes, optoelectronic devices, transistors, etc.

교과목표

- Understanding of solid-state physics related to electronic devices and semiconductor processes
- Fundamental principles of representative electronic device operation, and electrical characteristics of the same as circuit elements

주차별 강의계획

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학습윤리

1. It is unacceptable to copy others own ideas or words without properly crediting the sources. Any issue related to the plagiarism will be treated carefully, but thoroughly.
2. DO NOT distribute course materials without consent of the instructor.
3. Cheating is NOT allowed for all the course activities such as attendance, assignments, and exams.

교과정책(방침)

- This course is only for students in Department of Semiconductor Engineering.
 - “반도체 물리 및 소자” 수업을 이수한 학생만 수강 가능.
 - This course will follow the university rules (대구경북과학기술원 학칙) on academic courses, exams and grading.
1. ATTENDANCE POLICY: Up to two classes may be missed without a penalty. Beyond the two, there will be prorated penalty. Based on the DGIST rules, students must attend more than 75% of the entire classes in order not to fail (학칙 제48조).
 2. POLICIES ON MISSED EXAM: Students will follow the university rule (학칙 제49조).
 3. ACADEMIC INTEGRITY: Ethical behavior is an essential component of learning and scholarship. Students are expected to understand, and adhere to, the University's academic integrity policy. Please refer to the following resources: <http://curation.dgist.ac.kr/curation/w/370>